

CHAPTER 4

Using Covariates in Experimental Design and Analysis

One attractive feature of randomized experiments is that they generate unbiased estimates of the average treatment effect regardless of whether the researcher accounts for other causes of the outcome. The problem of *omitted variables* that bedevils nonexperimental research is addressed by random assignment. Randomly assigned treatments are statistically independent of all observed and unobserved variables. Any correlation between the treatment and other factors that affect outcomes arises purely by chance, due to the way in which observations happened to be allocated to treatment and control groups. Because the experimental researcher controls the assignment process, the sampling distribution of estimated ATEs can be studied in a rigorous manner, and precise inferences may be drawn when evaluating the sharp null hypothesis of no treatment effect. Experimentation makes causal inference possible even when the researcher has a very limited understanding of why some observations have different potential outcomes than others.

Although *covariates*, or supplementary variables that predict outcomes, are not required for unbiased estimation, they may be put to good use when available. This chapter covers several useful roles that covariates play in experimental design and analysis. We begin by discussing three ways in which covariates may be used in the analysis of experimental data. One technique is to use covariates to rescale the dependent variable so that potential outcomes have less variance, which in turn improves the precision with which treatment effects may be estimated. A second technique, regression analysis, uses covariates to eliminate observed differences between treatment and control groups and to reduce the variability in outcomes. The net effect is usually an improvement in the precision with which the treatment effect is estimated. Regression may also be used to check for data-handling errors that potentially undermine random assignment of observations to treatment and control groups. A third use of covariates is block randomized experimental design, which we introduced in Chapter 3. Based on intuitions about which covariates are likely to predict outcomes, the researcher forms a set of relatively homogenous groups, or blocks, each with

What you will learn from this chapter:

1. How to use covariates to obtain more precise estimates of the average treatment effect.
2. How covariates may be used to check the integrity of the random assignment procedure.
3. How block randomized experiments can improve precision, especially in small samples.

different expected outcomes. Random assignment to treatment and control is conducted separately within each block. The final part of the chapter weighs the pros and cons of incorporating covariates into experimental research. Because covariates may be used in many ways, they place additional discretion in the hands of the analyst, who may consciously or unconsciously make decisions that bias the results in a particular direction. Covariates are also a common source of error in the analysis of experimental data, either because researchers attach a causal interpretation to the covariates' apparent "effect" on the outcome or because researchers fail to account for block randomized assignment when analyzing experimental results. Appropriate use of covariates is best worked out during the design phase of an experiment, before the results become known and while the complications that arise from blocking are still fresh in mind.

4.1 Using Covariates to Rescale Outcomes

Perhaps the most intuitive way to begin a discussion of covariates is to consider the advantages of gathering what amounts to a pre-test prior to the launch of an experiment. For example, this pre-test might be an assessment of economic conditions prior to the implementation of an anti-poverty intervention, or of past voter participation in a study designed to assess whether an intervention increases voter turnout. The pre-test need not be a "test" per se but rather any observed variable (X_i) or set of variables (X_i) that is thought to predict potential outcomes.

One of the key assumptions associated with the pre-test and covariates more generally is that they are unaffected by treatment assignment. We need not use potential outcomes notation when describing covariates; it is assumed that the schedule of covariates remains fixed regardless of whether an observation is assigned to treatment or control. The assumption that X_i is fixed casts suspicion on so-called

post-treatment covariates. Post-treatment covariates are measured after the intervention and are therefore potentially affected by the experimental assignment. Granted, characteristics such as sex or race may be so stable that we may safely assume that they are unaffected by the intervention even if they are measured afterwards. But this is a substantive assumption that will need to be evaluated on a case-by-case basis. For purposes of exposition, we will focus solely on *pre-treatment* covariates. The values of X_i are assumed to be fixed constants that are observed prior to the random assignment of subjects to treatment and control.

If we allocate subjects randomly with equal probability to treatment and control groups, the expected value of X_i in the treatment group will be the same as the expected value of X_i in the control group.

$$E[X_i] = E[X_i | D_i = 1] = E[X_i | D_i = 0], \quad (4.1)$$

where D_i refers to whether a hypothetical assignment causes a subject to receive the treatment. This equality has a number of useful implications. It is the basis for difference-in-differences estimation, which we discuss next, and randomization checks, which we discuss in section 4.3.

What if we were to redefine the outcome measure as the change from pre-test to post-test? Suppose that Y_i represents the post-test, but instead of comparing the average value of Y_i for $d_i = 1$ to the average value of Y_i for $d_i = 0$ we were to compare the average value of $(Y_i - X_i)$ for $d_i = 1$ to the average value of $(Y_i - X_i)$ for $d_i = 0$. The analysis of change scores is called the *difference-in-differences* estimator. Does the difference-in-differences estimator produce unbiased results when treatments are randomly assigned? In order to answer this question, let's consider the expected value of the difference-in-differences estimator in light of the equality in equation (4.1):

$$\begin{aligned} E(\widehat{ATE}) &= E[Y_i - X_i | D_i = 1] - E[Y_i - X_i | D_i = 0] \\ &= E[Y_i | D_i = 1] - E[X_i | D_i = 1] - E[Y_i | D_i = 0] + E[X_i | D_i = 0] \\ &= E[Y_i(1)] - E[Y_i(0)]. \end{aligned} \quad (4.2)$$

The last line is the same expectation as the conventional difference-in-means estimator discussed in Chapter 2. So estimating the ATE using change scores provides another way to obtain unbiased estimates from experimental data.

The two estimators, difference-in-means and difference-in-differences, both generate unbiased estimates. In any finite sample, the two approaches may generate different estimates due to chance. In order to figure out which estimate is more trustworthy, we must figure out which estimation approach is less prone to sampling variability.

In the last chapter, we presented equation (3.4), the formula for the standard error of the difference-in-means estimator. For ease of exposition, let's assume that

an experiment places half of the observations into the treatment group such that $m = N/2$. Under this experimental design, equation (3.4) simplifies to:

$$SE(\widehat{ATE}) = \sqrt{\frac{\text{Var}(Y_i(0)) + \text{Var}(Y_i(1)) + 2\text{Cov}(Y_i(0), Y_i(1))}{N-1}}. \quad [4.3]$$

We can adapt this formula to evaluate the relative merits of the difference-in-differences estimator, simply replacing Y_i with $(Y_i - X_i)$. We replace $\text{Var}(Y_i(0))$ with $\text{Var}(Y_i(0) - X_i) = \text{Var}(Y_i(0)) + \text{Var}(X_i) - 2\text{Cov}(Y_i(0), X_i)$ and $\text{Var}(Y_i(1))$ with $\text{Var}(Y_i(1) - X_i) = \text{Var}(Y_i(1)) + \text{Var}(X_i) - 2\text{Cov}(Y_i(1), X_i)$ to obtain:

$$\begin{aligned} SE(\widehat{ATE}') &= \sqrt{\frac{\text{Var}(Y_i(0) - X_i) + \text{Var}(Y_i(1) - X_i) + 2\text{Cov}(Y_i(0) - X_i, Y_i(1) - X_i)}{N-1}} \\ &= \sqrt{\frac{\text{Var}(Y_i(0)) + \text{Var}(Y_i(1)) + 2\text{Cov}(Y_i(0), Y_i(1)) + 4(\text{Var}(X_i) - \text{Cov}(Y_i(0), X_i) - \text{Cov}(Y_i(1), X_i))}{N-1}}. \end{aligned} \quad [4.4]$$

Comparing equations (4.3) and (4.4), we see that the difference-in-differences estimator has a smaller sampling variance than the difference-in-means estimator when:

$$\text{Cov}(Y_i(0), X_i) + \text{Cov}(Y_i(1), X_i) > \text{Var}(X_i) \quad [4.5]$$

or

$$\frac{\text{Cov}(Y_i(0), X_i)}{\text{Var}(X_i)} + \frac{\text{Cov}(Y_i(1), X_i)}{\text{Var}(X_i)} > 1. \quad [4.6]$$

In other words, if we were to regress $Y_i(1)$ on X_i and regress $Y_i(0)$ on X_i , the sum of the slope coefficients would have to exceed 1 in order for the rescaling approach to produce more efficient estimates of the ATE.¹ We cannot literally perform these regressions because we lack the full schedule of potential outcomes for all observations, but we can approximate them based on the Y_i that we do observe. The bottom line is that when a covariate strongly predicts potential outcomes, difference scores can produce substantial gains in precision.

In practical terms, this result implies that researchers should take advantage of opportunities to gather background information that may be useful in predicting potential outcomes. Conducting a pre-test or other background survey is not necessary for unbiased estimation, but it can pay dividends in terms of the precision with which the ATE is estimated. Sometimes researchers deliberately focus their interventions on subjects for whom background information is readily available.

1 The connection between equation (4.6) and regression is that the regression coefficient one obtains when regressing Y on X is $\text{Cov}(Y, X)/\text{Var}(X)$.

Researchers may confront trade-offs when considering whether to gather pre-test information. Budget constraints may limit how much data a researcher can collect on subjects prior to the implementation of a treatment. For example, in many cases, researchers must conduct household surveys in order to obtain data on covariates, in which case budget constraints may force a choice between gathering additional subjects and gathering more extensive background information for each subject.

A more serious concern arises when the administration of a pre-test changes the way that participants respond to the treatment and to subsequent attempts to measure outcomes. Suppose the outcome measure is the opinions that subjects express on a survey at the end of an experiment. The administration of the pre-test may suggest to participants that there is a socially desirable way to respond to outcome measures thereafter. Although pre-tests may improve precision, precision takes a backseat to more important concerns, such as the threats that pre-testing may present to core assumptions like excludability. Bias may be introduced if the pre-test provokes different reactions in the treatment and control groups. For example, the control group may shrug off a pre-test because no intervention follows; the treatment group on the other hand may interpret the pre-test as a signal that certain outcomes are expected to change in the wake of the treatment. If this expectation affects outcomes, the experiment measures a combination of the causal effect of the treatment and the causal effect of taking the pre-test. If pre-testing jeopardizes a study's ability to estimate the ATE of interest, researchers should look to other prognostic covariates that can be obtained less obtrusively.

In order to illustrate the potential statistical value of covariates, we present a stylized example inspired by Muralidharan and Sundararaman's study of the educational effects of providing teachers in India with financial bonuses according to their students' academic success.² For our purposes, imagine 40 rural primary schools, 20 of which are assigned to the treatment condition: bonus pay for teachers. Academic achievement in each school is assessed by means of a pre-test and a post-test; the pre-test is administered prior to the start of the school year, and the post-test is administered at the end of the school year. For simplicity, assume that the same set of students takes both tests. Finally, in order to illustrate the sampling distributions of different estimators, suppose we know the full schedule of potential outcomes associated with each school.

The schedule of potential outcomes for the 40 schools is presented in Table 4.1. In keeping with the Muralidharan and Sundararaman experiment, our data incorporate a treatment effect of approximately one-quarter of a standard deviation: the treatment effect is 4.0, while the standard deviation in $Y_i(0)$ is 15.5. Our simulation also presupposes a high degree of persistence in scores over the course of a year,

2 Muralidharan and Sundararaman 2011.

TABLE 4.1

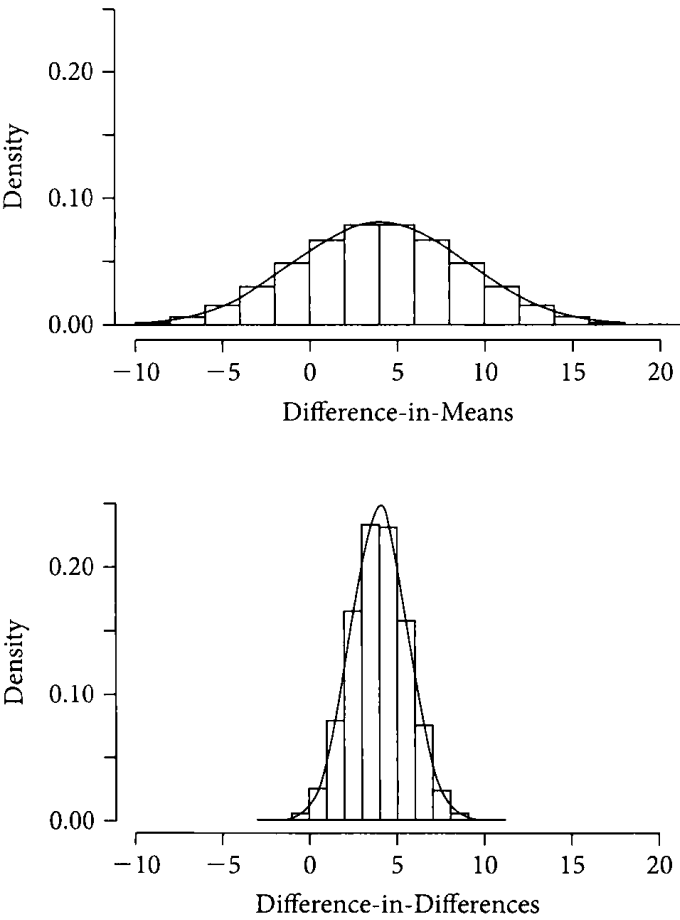
Schedule of potential outcomes, pretest scores, and group assignments for simulated teacher incentives experiment

Observation	$Y_i(1)$	$Y_i(0)$	d_i	X_i	X_weak_i
1	5	5	0	6	25
2	15	5	1	8	12
3	12	6	1	5	25
4	19	9	0	13	27
5	17	10	0	9	10
6	18	11	0	15	24
7	24	12	0	16	21
8	11	13	0	17	25
9	16	14	0	19	35
10	25	19	1	23	28
11	18	20	1	28	41
12	21	20	0	28	38
13	17	20	0	9	30
14	24	21	1	16	20
15	27	24	1	23	24
16	26	25	0	15	26
17	30	27	1	23	22
18	37	27	0	33	34
19	43	30	1	42	37
20	39	32	0	31	21
21	36	32	0	29	40
22	27	32	0	28	34
23	33	32	1	35	36
24	37	35	1	28	37
25	48	35	0	41	48
26	39	37	1	37	46
27	42	38	1	32	25
28	37	38	1	37	21
29	53	41	0	36	19
30	50	42	1	44	44
31	51	43	1	48	50
32	43	44	1	43	48
33	55	45	1	55	46
34	49	47	0	53	47
35	48	48	0	51	47
36	52	51	1	43	39
37	59	52	0	57	50
38	52	52	0	51	46
39	55	57	1	49	54
40	63	62	1	55	42

with the pre-test (X_i) predicting 87% of the variance in the $Y_i(0)$ scores.³ (Later, we consider the consequences of administering a less reliable version of the pre-test that predicts only 50% of the variance in $Y_i(0)$.) The pre-test is also scaled in roughly the same way as the outcome measures: $Y_i(0)$, $Y_i(1)$, and X_i all have standard deviations between 15 and 16.

In order to examine the sampling distribution of the difference-in-means estimator, we simulated 100,000 experiments in which 20 of the 40 observations were assigned to the treatment group. The average estimate of the ATE across the 100,000 experiments was 4.0, with a standard deviation of 4.768. This empirically derived standard error is very close to the 4.773 that we obtain by applying the exact standard error formula in equation (3.4). The top panel of Figure 4.1 shows that the sampling distribution is bell-shaped with a mean of 4, but it is so widely dispersed that 20.4% of all simulated experiments produce negative ATE estimates.

FIGURE 4.1
Sampling distribution of two estimators: difference-in-means
and difference-in-differences



3 This number is the R^2 from a regression of $Y_i(0)$ on X_i using the data in Table 4.1.

By contrast, rescaling the outcome to reflect the change from pre-test to post-test—the difference-in-differences estimator—produces estimates of the ATE that are both unbiased and precise. Based on 100,000 simulated experiments, the bottom panel of Figure 4.1 shows the average estimate to be 4.0 with a standard deviation of 1.53. Just 0.4% of the simulated experiments generate negative ATE estimates. In order to put this improvement into perspective, consider the increase in sample size required to bring about a $1 - (1.53/4.77) = 68\%$ reduction in standard error. One would need to expand the sample by a factor of $1/(1 - 0.68)^2 = 9.8$, which implies expanding the experiment from 40 schools to 392.

Not all experimenters have the luxury of working with highly predictive pre-tests. In the actual Muralidharan and Sundararaman study, the pre-test predicts about half of the variance in the control group's outcomes. A pre-test's predictive accuracy may be diminished by several factors: (1) the pre-test may contain measurement error; (2) the trait that one seeks to measure may change between the administration of the pre-test and the measurement of outcomes, and, in particular, subjects may change at different rates; or (3) the outcomes may be measured with error. Here, we illustrate the first of these possibilities by considering a less reliable version of the pre-test that predicts 50% of the variance in $Y_i(0)$, labeled X_weak_i in Table 4.1. Although this noisy version of the pre-test is less than ideal, it nevertheless improves precision.⁴ Rescaling the outcome by this pre-test generates a standard error of 3.48. When compared to the simple difference-in-means estimator, the $1 - (3.48/4.77) = 27\%$ reduction in standard error is considerable. Even the noisy pre-test offers gains in precision equivalent to an 88% increase in sample size.

4.2 Adjusting for Covariates Using Regression

Regression is a flexible tool for analyzing experimental data. Regression allows the researcher to adjust for many covariates simultaneously. For example, if researchers were to administer a pre-test and also gather data on the socioeconomic characteristics of each school, regression could estimate the ATE controlling for both of these background variables. Regression may also be used to estimate the effects of several different treatments, or treatments that are assigned in different dosages (e.g., 100, 200, or 300 hours of instruction), a topic that we revisit in Chapters 9 and 11.

In order to see the connection between regression and the potential outcomes framework, let's rewrite the potential outcome equation (2.2) as a linear regression equation, ignoring covariates for the moment:

4 The noisy pre-test easily satisfies the requirement of equation (4.6). A regression of $Y_i(1)$ on the pre-test gives a coefficient of 0.87, and a regression of $Y_i(0)$ on the pre-test gives a coefficient of 0.95. The sum of the coefficients is therefore greater than 1.0.

$$\begin{aligned}
 Y_i &= Y_i(0)(1 - d_i) + Y_i(1)d_i = Y_i(0) + (Y_i(1) - Y_i(0))d_i = \underbrace{\mu_{Y(0)}}_{\text{Intercept}} \\
 &+ \underbrace{[\mu_{Y(1)} - \mu_{Y(0)}]d_i}_{\text{ATE}} + \underbrace{Y_i(0) - \mu_{Y(0)} + [(Y_i(1) - \mu_{Y(1)}) - (Y_i(0) - \mu_{Y(0)})]d_i}_{\text{Disturbance}} \\
 &= a + bd_i + u_i.
 \end{aligned} \tag{4.7}$$

Here, the intercept a is equal to $\mu_{Y(0)}$, or the average value of the untreated potential outcomes for all N subjects. The slope b is equal to $\mu_{Y(1)} - \mu_{Y(0)}$ and represents the shift in average potential outcomes, or the ATE. The independent variable is the treatment, d_i . The unobserved part of the regression model is the disturbance term $u_i = Y_i(0) - \mu_{Y(0)} + [(Y_i(1) - \mu_{Y(1)}) - (Y_i(0) - \mu_{Y(0)})]d_i$. The disturbance term comprises the idiosyncratic variation in untreated responses $Y_i(0) - \mu_{Y(0)}$, plus the idiosyncratic variation in treatment effects $[(Y_i(1) - \mu_{Y(1)}) - (Y_i(0) - \mu_{Y(0)})]d_i$. The disturbance term is actually much simpler than it appears. When an observation is assigned to the control group, $d_i = 0$, and the disturbance term equals $Y_i(0) - \mu_{Y(0)}$; when an observation is assigned to the treatment group, $d_i = 1$, and the disturbance term equals $Y_i(1) - \mu_{Y(1)}$.

When ordinary least squares (OLS) regression is used to estimate the parameters of a regression model like equation (4.7), the estimates of the intercept ($\hat{a}$) and slope ($\hat{b}$) are chosen so as to minimize the sum of squared residuals, where a residual is defined as $Y_i - (\hat{a} + \hat{b}d_i)$. A regression of Y_i on d_i produces exactly the same estimate of the ATE as a simple difference-in-means.⁵

In the previous section, we rescaled the dependent variable so that the outcome was defined as the change from pre-test to post-test. In order to see how rescaling looks when expressed as a regression equation, we rewrite the dependent variable as $Y_i^* = Y_i - X_i$:

$$\begin{aligned}
 Y_i^* &= Y_i - X_i = Y_i(0)(1 - d_i) + Y_i(1)d_i - X_i \\
 &= a + bd_i + u_i - X_i = a + bd_i + u_i^*.
 \end{aligned} \tag{4.8}$$

The last equality reflects the fact that the new disturbance term is $u_i^* = u_i - X_i$.

⁵ Regression may also be used to estimate the standard error of the treatment effect. When the treatment and control groups contain the same number of subjects, the regression standard error is identical to the estimator in equation (3.6). When the treatment and control groups are of different sizes, regression with “robust” standard errors approximates the estimator in equation (3.6). See Samii and Aronow 2012. When regression is applied to cluster randomized experiments, the standard error may be estimated using “robust cluster” standard errors, which allow disturbances within clusters to be related but assume that disturbances from different clusters are independent. This method of estimating standard errors may be unreliable when the data contain fewer than 20 clusters but becomes increasingly accurate as the number of clusters increases. See Angrist and Pischke 2009, section 8.2.3.

An alternative to rescaling the outcome is to expand the regression model by including one or more covariates as right-hand-side predictors, an approach called *regression adjustment* or *covariate adjustment*. What was before a regression of Y_i on d_i now becomes a regression of Y_i on d_i and X_i . The regression model may be written as follows:

$$Y_i = Y_i(0)(1 - d_i) + Y_i(1)d_i = a + bd_i + cX_i + (u_i - cX_i), \quad [4.9]$$

where the disturbance is the last term in parentheses. This regression adjustment model closely resembles the rescaled regression model presented in equation (4.8), except that now we have included a parameter c in the model whereas equation (4.8) assumed that $c = 1$. If the regression estimate of c is close to 1, regression adjustment and rescaled regression will generate very similar estimates of b .

It should be stressed that the regression estimate $\hat{c}$ has no causal interpretation. The values of Y_i are unaffected by X_i ; the middle part of equation (4.9) underscores this point by reiterating that Y_i is solely a function of potential outcomes and treatment status. X_i plays no role in the causal process—using an accounting identity, we added it to the regression model and subtracted it from the disturbance term. The only reason to include X_i as a regressor is that doing so subtracts cX_i from the disturbance. When X_i predicts outcomes, this subtraction reduces the amount of unexplained variation in Y_i , which in turn reduces the standard error of $\hat{b}$. In other words, the decision to include X_i in the regression model is based on whether it is thought to *predict* outcomes; we remain agnostic about whether X_i *affects* outcomes.

What is the advantage of using covariates as regressors? The answer is flexibility. We can include several covariates as right-hand-side variables and let OLS determine their respective scaling parameters. In principle, this approach reduces disturbance variability more effectively than rescaled regression, enabling us to estimate b more precisely.⁶

Are there drawbacks to using regression to control for covariates? In small samples ($N < 20$), controlling for covariates can lead to bias,⁷ and rescaling the outcome is probably a safer strategy than regression adjustment. In larger samples ($N \geq 20$), the inclusion of a single covariate typically leads to negligible bias. To avoid bias, regression models that include multiple covariates should have at least 20 more observations than covariates. Whether the inclusion of covariates leads to a reduction in standard error depends on the application. Prognostic covariates may reduce

6 In the schooling example, covariate adjustment using regression produces no gains in precision compared to rescaled regression because the value of c is close to 1.0. Simulations show the standard error of $\hat{b}$ is 1.53 with the rescaled outcomes estimator, as opposed to 1.54 for the regression adjusting for the pre-test as a covariate.

7 Freedman 2008. This bias diminishes rapidly as sample size increases; see Green and Aronow 2011. In samples of any size, regression adjustment does not produce bias when the treatment and control groups are of equal size and treatment effects are constant. Thus, for balanced designs, regression adjustment is unbiased under the sharp null hypothesis of no effect for any subject.

disturbance variability and improve the precision with which treatment effects are estimated. When applying regression to large samples, researchers should include any pre-treatment covariate that previous research (including observational research), pilot testing, or theoretical intuition suggests will predict outcomes. Including covariates that are weakly related to outcomes does little or nothing to reduce sampling variability.⁸ However, so long as the researcher does not use the experimental outcomes (the observed values of Y_i) when deciding which covariates to include, the statistical drawbacks of adjusting for covariates are fairly minor, particularly when the experiment involves a large number of subjects.

More serious concerns arise when researchers consider experimental outcomes when deciding which covariates to include in their regression models. For example, after outcome data become available, a researcher might explore a variety of regression models that use different combinations of covariates to predict Y_i . This type of analysis introduces an element of discretion in terms of what results are reported. Perhaps unconsciously, the researcher may settle on a regression model that makes the estimated ATE look impressive or interesting, a decision rule that jeopardizes the unbiasedness of the estimator.

In order to dampen the role of discretion, researchers are encouraged to plan their covariate adjustment procedures in advance of the experiment. By following a predetermined plan, researchers remove the temptation to allow observed outcomes to guide their modeling choices. If the plan specifies a list of what are believed to be prognostic covariates, the covariate-adjusted estimate of the ATE should be regarded as the best guess of the true ATE, because it was expected to be more precise than the difference-in-means estimate. Regardless of whether the analysis is planned in advance or not, researchers should present difference-in-means estimates alongside covariate-adjusted estimates so that readers can judge whether the inclusion of covariates is consequential.

4.3 Covariate Imbalance and the Detection of Administrative Errors

The random assignment of any finite number of observations inevitably produces some degree of *imbalance*, or correlation between the assigned treatment and one or more covariates. By controlling for covariates, whether through rescaling the dependent variable or through regression adjustment, the researcher reestablishes balance.

⁸ It is possible to construct examples in finite samples where the inclusion of covariates helps or hurts precision, but Lin (2010) shows asymptotically (i.e., as sample size increases) that regression adjustment produces more precise estimates than a simple difference-in-means when similar numbers of subjects are in treatment and control or when treatment effects are uncorrelated with the covariate. When $N \neq 2m$ and treatment effects are correlated with the covariate, regression adjustment may lead to improved precision if the covariate strongly predicts outcomes.

BOX 4.1**Definition: Indicator or “Dummy” Variables**

A categorical variable can be recoded into a series of indicator variables, also known as dummy variables. For example, consider a variable defining three experimental blocks {A, B, C}. The three blocks become three variables: Variable 1 is coded 1 if the block variable equals A and is coded 0 otherwise. Variable 2 is coded 1 if the block variable equals B and is coded 0 otherwise. Variable 3 is coded 1 if the block variable equals C and is coded 0 otherwise. When using indicator variables as covariates in a regression, include all but one of the indicator variables. In this example, any two of the three dummy variables may be used.

For example, if one estimates a regression like equation (4.9), the regression will eliminate any linear relationship between the assigned treatment and the pre-test. If one wanted to eliminate both linear and nonlinear relationships between the treatment and the covariate while allowing for possibly nonlinear relationships between the covariate and the outcome, one could either control for a polynomial function of the pre-test (such as X_i , X_i^2 , and X_i^3) or code the pre-test as a series of indicator variables in order to examine treatment effects among those who share the same pre-test score. (See Box 4.1.) The broader point is that imbalance on measured indicators is a tractable problem that can be handled by methods such as regression analysis.

Imbalance is a cause for concern not because it presents an insurmountable estimation problem, but because it may call into question the soundness of the random assignment procedure or the administrative handling of the data. Suppose one's randomization of the schooling experiment were to allocate to the treatment group all 20 of the highest-performing schools as gauged by the pre-test. To be sure, such a randomization is conceivable as one of the 137,846,528,820 possible randomizations. On the other hand, it could be a symptom of some sort of clerical error. Maybe someone forgot to randomize the list of schools and left it sorted according to pretest scores? Or maybe there was a miscommunication about which schools were randomly assigned to the treatment?

When concerns like these arise, the first thing to do is retrace the steps that were taken when the observations were randomly assigned. Does the dataset contain the random numbers on which the random assignment was based? If so, do the treatment assignments appear to follow the randomly specified schedule? Is there a computer program that reproduces the random numbers and the process by which they were used to guide the allocation of subjects? These kinds of internal audits may solve the

mystery by reference to the randomization procedure itself or other administrative problems that caused some observations to be excluded, double-counted, or miscoded.

Because administrative errors are possible whenever one generates primary data, the nominal standard errors associated with experiments potentially understate the true degree of uncertainty surrounding an experimental result. For that reason, experimenters should carefully investigate the soundness of the randomization procedure, especially if the randomization was conducted by a third party, such as a government agency. When reporting experimental results, researchers are expected to give a step-by-step account of the random assignment process and to describe in detail any instances in which subjects are lost or excluded after allocation (see Chapter 13).

One way to bolster confidence in the integrity of the randomization procedure and the administrative care with which the data were handled is to provide a statistical description of the balance between treatment and control groups on a range of available covariates. Often, one of the first tables in a research article reports the distribution of pre-test scores and other covariates for each of the experimental groups. This table enables the reader to assess two things. First, do prognostic variables, such as pre-test scores, differ appreciably between treatment and control groups? If so, this pattern will help the reader understand why the estimated ATE varies depending on whether covariates are included. Second, for the entire set of covariates, are the imbalances larger than one would expect from chance alone? This question can be answered by means of a statistical test and accompanying simulation. For binary treatments, the statistical test involves a regression of the assigned treatment on all of the covariates and calculation of the F-statistic.⁹ In order to calculate the p -value of the F-statistic, one uses randomization inference.¹⁰ As explained in section 3.4, randomization inference involves simulating the random assignment procedure many times, calculating the test statistic for each sample. When all of the possible randomizations have been simulated,¹¹ the collection of test statistics represents the exact

⁹ Alternatively, one may use logistic regression and calculate the log-likelihood statistic. For experiments with many treatments, the same regression is performed using multinomial logistic regression, and again one calculates a log-likelihood statistic.

¹⁰ The F-statistic is used to evaluate the goodness of fit of a regression model. In this case, the statistic is used to test the null hypothesis that the covariates predict the treatment assignments no better than would be expected by chance. Under the null hypothesis, the only predictor of the outcome is the intercept. Under the alternative hypothesis, the q covariates do in fact systematically predict treatment assignments. Call SSR_1 the sum of squared residuals obtained when regressing treatment assignments on an intercept. Call SSR_2 the sum of squared residuals obtained when regressing treatment assignments on an intercept and the q covariates. The F-statistic is

$$\frac{SSR_1 - SSR_2}{q} \bigg/ \frac{SSR_2}{N - q - 1}.$$

¹¹ When the number of possible randomizations becomes too large to compute in a timely fashion, one can simulate a large number of random assignments (e.g., 100,000) and treat these as a random sample from the set of possible assignments.

sampling distribution under the null hypothesis that no covariates have any effect on the assigned treatment, which is implied by random assignment. One obtains the p -value of the actual experimental sample's test statistic by finding its location in the sampling distribution of all test statistics. For example, if 43% of the test statistics are at least as large as the observed test statistic, the p -value is 0.43. This procedure works well regardless of the number of covariates, whereas more conventional significance tests break down when the number of covariates is large.¹² Another advantage of this procedure is that it may be used for any randomized design, whether simple, blocked, or clustered.

Suppose this testing procedure reveals that imbalance is greater than one would expect by chance—the p -value is 0.01. This result should prompt a thorough review of the random assignment procedure and any possible data-handling mistakes. If these procedures are checked and found to be satisfactory, proceed on the assumption that the imbalance is due to random chance. One-in-a-hundred outcomes do occur, and if you conduct enough experiments, you can expect to encounter this kind of random anomaly. The imbalance test should be reported, and estimation results should be presented with and without covariate adjustment.

Does random imbalance on measured covariates mean that the experimental estimates are biased? Bias is a property of an estimation procedure, not a specific estimate. The procedure of randomly assigning observations to treatment and control generates unbiased estimates of the ATE, even though some estimates are too high or too low and even though some samples display covariate imbalance. Ex ante (that is, before we conduct our random assignment), we expect our experimental procedure to render, on average, the ATE.

After conducting the randomization and noticing imbalance on one or more prognostic covariates (but before observing outcomes), this expectation may change. For example, suppose we notice that our random assignment has produced a treatment group whose average pre-test scores are higher than the average pre-test scores in the control group. Ex post, we might now expect our estimated ATE to be a higher-than-average draw from the sampling distribution. The teacher incentives study illustrates how these expectations play out numerically. The difference-in-means provides an unbiased estimate of the true ATE, which in this case is 4.0. However, if we restrict our attention to only those randomizations in which the average pre-test scores of the treatment group exceeded the average pre-test scores of the control group, the difference-in-means estimator generates an average estimate of 7.7. Conditional on observing an imbalance on a prognostic variable that favors the treatment group, $E(\widehat{ATE}) > ATE$.

Surprisingly, it is possible to recover the ATE even from this pathological subset of experiments. Despite the fact that these randomizations seem to favor the treatment

¹² Hansen and Bowers 2008.

group, when we control for pre-test scores using regression, these imbalanced experiments nevertheless yield unbiased estimates: the average estimated ATE is 4.0. The bottom line is that if a covariate is imbalanced due to random assignment, controlling for this imbalance produces unbiased estimates.¹³ In practice, if you find evidence of imbalance in a continuous covariate, you may wish to control for this variable using a flexible functional form. For example, if you were to find imbalance on a pre-test score, you might want to control for the pre-test score as well as squared values of the pre-test score.

Sometimes researchers worry that observed imbalance is symptomatic of a broader problem of imbalance among other unobserved factors that affect outcomes. So long as imbalance is solely due to random chance (as opposed to administrative error) and so long as we control for the covariate that is imbalanced, there is no reason to expect imbalance on other covariates or on unmeasured causes of the outcome variable. For example, imagine that poorer villages tend to have lower pre-test scores. Suppose the treatment group has higher pre-test scores on average. By controlling for pre-test scores, we render irrelevant the correlation between wealth and test scores; in effect, we are comparing treatment and control outcomes within schools that have the same pre-test score. When random imbalance turns up in a covariate that is thought to predict outcomes, the solution is to control for that covariate.

4.4 Blocked Randomization and Covariate Adjustment

Random assignment sometimes puts researchers in a quandary. Chance imbalances cause the estimated treatment effect to move around depending on whether and how covariates are controlled. Sometimes the introduction of covariates leads to a decisive improvement in precision, but other times the estimated standard errors are similar, and the choice between the difference-in-means estimator and estimators that use covariate adjustment is less clear cut. The situation becomes especially murky in the absence of a planning document that, prior to the collection of experimental outcomes, lays out the ways in which the researcher intends to analyze the data. The persuasiveness of experimental evidence diminishes when results depend on the analyst's post hoc choices.

Blocked random assignment is a way to prevent this conundrum. Instead of randomly assigning all observations at once, the researcher divides the observations into

¹³ Regression is subject to the small-sample bias described in section 4.2, but this bias is negligible in this example because the N is fairly large. Note that the benefits of controlling for imbalance do not necessarily hold when imbalance is caused by systematic factors, such as attrition that disproportionately affects one experimental group. See Chapter 7.

more homogeneous blocks and randomly allocates observations within each block. Blocking ensures that all variables used to create strata will be balanced and tends to improve the precision with which the ATE is estimated. For this reason, experimenters tend to follow the dictum “Block what you can, and randomize what you cannot.”

With ample preparation time, a researcher can block on several variables simultaneously. For example, a researcher could divide schools according to religion and average levels of poverty. Within each category, observations may be randomly allocated to treatment and control, so that each religion/poverty category allocates $100 * (m/N)\%$ of its observations to the treatment group. If the number of observations is small, the dataset may not contain more than one observation with the same covariate profile, in which case the researcher may block on the most prognostic covariate or an index that summarizes the covariates into a single prognostic score.

Unfortunately, deciding which variables to use in order to construct blocks involves some guesswork. Blocking produces the greatest gains in precision when the variables used to form blocks strongly predict outcomes, but outcomes obviously cannot be observed until the experiment is actually conducted. Researchers therefore look to other studies, including observational studies, for clues about which covariates are the strongest predictors. For example, when planning an experiment that aims to increase elementary students’ test scores, one could look at prior research on how these test scores are predicted by region, ethnicity, and scores in prior years. Exercise 4.6 illustrates how one might analyze data for hints about how to form blocks.

Blocking on prognostic covariates helps improve the precision with which treatment effects are estimated. In Chapter 3, we compared the sampling distribution under block random assignment and complete random assignment. We found that when prognostic variables are used to form blocks, block randomization provides

BOX 4.2

Software Routines That Automate Blocking

Many software packages allow researchers to block observations prior to random assignment. One of these packages, *blockTools* (Moore 2010), is freely available as a package using the R software environment. The *blockTools* package uses a matching algorithm to automatically group units with similar covariate profiles. It can perform pair-matching or construct blocks of any size, so long as all blocks contain the same number of observations. *blockTools* is also capable of implementing more complex randomization designs.

An annotated example of blocking in R may be found at <http://isps.research.yale.edu/FEDAI>. See exercise 4.6.

more precise estimates of the ATE. What happens when we compare block randomization and complete randomization but use regression to control for the variables used to form blocks? For simplicity, suppose subjects in all blocks have the same probability of being assigned the treatment. Blocking eliminates correlation between the assigned treatment and the variables used to form blocks. Returning to the teachers' incentives example in section 4.1, if 75% of the schools in both treatment and control groups are located in predominantly Hindu villages, the correlation between religion and assignment is exactly zero. Because the correlation is zero, regression produces the same estimated ATE regardless of whether we control for the variables used to form blocks.¹⁴ In other words, we obtain the same estimated ATE whether we regress outcomes on d_i or on both d_i and the covariate(s) used to form the blocks. On the other hand, if complete randomization is used, the covariates may be correlated with the treatment; simply by chance, the proportion of predominantly Hindu villages in the treatment group may be higher or lower than 75%. Regression will eliminate this correlation, but the price of doing so is known as the *collinearity penalty*.¹⁵ The larger the R^2 when d_i is regressed on the covariates used to form blocks, the more sampling variability there will be when regression is used to estimate the ATE.

The collinearity penalty tends to decline as the number of subjects increases. In large samples, it is rare to see an appreciable correlation between the assigned treatment and the covariates that could be used to form blocks. For example, suppose we were to conduct an experiment on 75 predominantly Hindu villages and 25 predominantly non-Hindu villages. If half of the 100 villages are randomly selected for treatment, there is a 95% chance that our treatment group will contain between 66% and 84% predominantly Hindu villages. The average R^2 when d_i is regressed on an indicator for village ethnicity is 0.01. Now suppose that the sample were much larger, 750 predominantly Hindu villages and 250 non-Hindu villages. If half of the 1,000 villages are assigned to treatment, there is a 95% chance that our treatment group will contain between 72% and 78% predominantly Hindu villages. The average R^2 when d_i is regressed on an indicator for village ethnicity is now just 0.001. In large samples, the sampling distribution under blocking is very similar to the sampling distribution under regression adjustment. Blocking merely guards against rare instances in which random chance causes covariates to be correlated with d_i .

Does blocking have a downside? The worst-case scenario occurs when blocks are formed using a covariate that fails to predict experimental outcomes. Fortunately,

14 In small samples, it may be impossible to form blocks such that the covariates are completely orthogonal to treatment assignment. In such cases, including the covariates used to form blocks may alter the estimated ATE, although the difference between the estimated ATE with and without covariate adjustment tends to be negligible.

15 The term collinearity refers to the R^2 when d_i is regressed on all of the covariates. If this R^2 were 1.0, regression would not be able to estimate the ATE because no variance would remain in d_i after removing the covariance that d_i shares with the covariates.

even in this case, the estimates remain unbiased and have no more sampling variability than the estimates one would have obtained from a complete randomization.¹⁶ The bottom line is that researchers should block whenever they have reason to believe that a covariate predicts potential outcomes. If a second or third covariate helps predict outcomes, create more refined blocks.

In order to illustrate the potential statistical advantages of blocking, we return to the schools example. In section 4.1, we saw that under complete random assignment, the difference-in-means estimator generated unbiased estimates with a standard error of 4.77. Now, we consider the sampling distribution when schools are blocked according to pre-test scores using the matched pair design described in Chapter 3, section 3.6.1.1. Blocks are created by sorting the 40 schools by pre-test scores in ascending order and forming pairs. The first two schools form the first pair; schools 3 and 4 form the second pair; and so forth. Within each pair, a coin flip determines which of the two schools is placed into the treatment group. This procedure ensures that pre-test scores bear no more than a trivial correlation with treatment assignment.

As noted in Chapter 3, estimation of the ATE involves computing a weighted average of the $\widehat{ATE}_j$ within each block. When the probability of assignment to the treatment condition is the same for every block, as in our matched pair example, regression gives identical results and makes estimation convenient. In order to estimate the ATE and its standard error, apply regression to the following model:

$$Y_i = a + bd_i + c_1X_{1i} + c_2X_{2i} + \dots + c_{19}X_{19i} + u_i, \quad (4.10)$$

where the 19 indicator variables X_{1i} through X_{19i} mark each of the blocking pairs. For example, X_{1i} is scored 1 when the observations belong to the first two schools on the sorted list, and 0 otherwise. The indicator variable marking the twentieth pair of observations is omitted from the model because the intercept for this omitted pair is represented by a , the constant term.

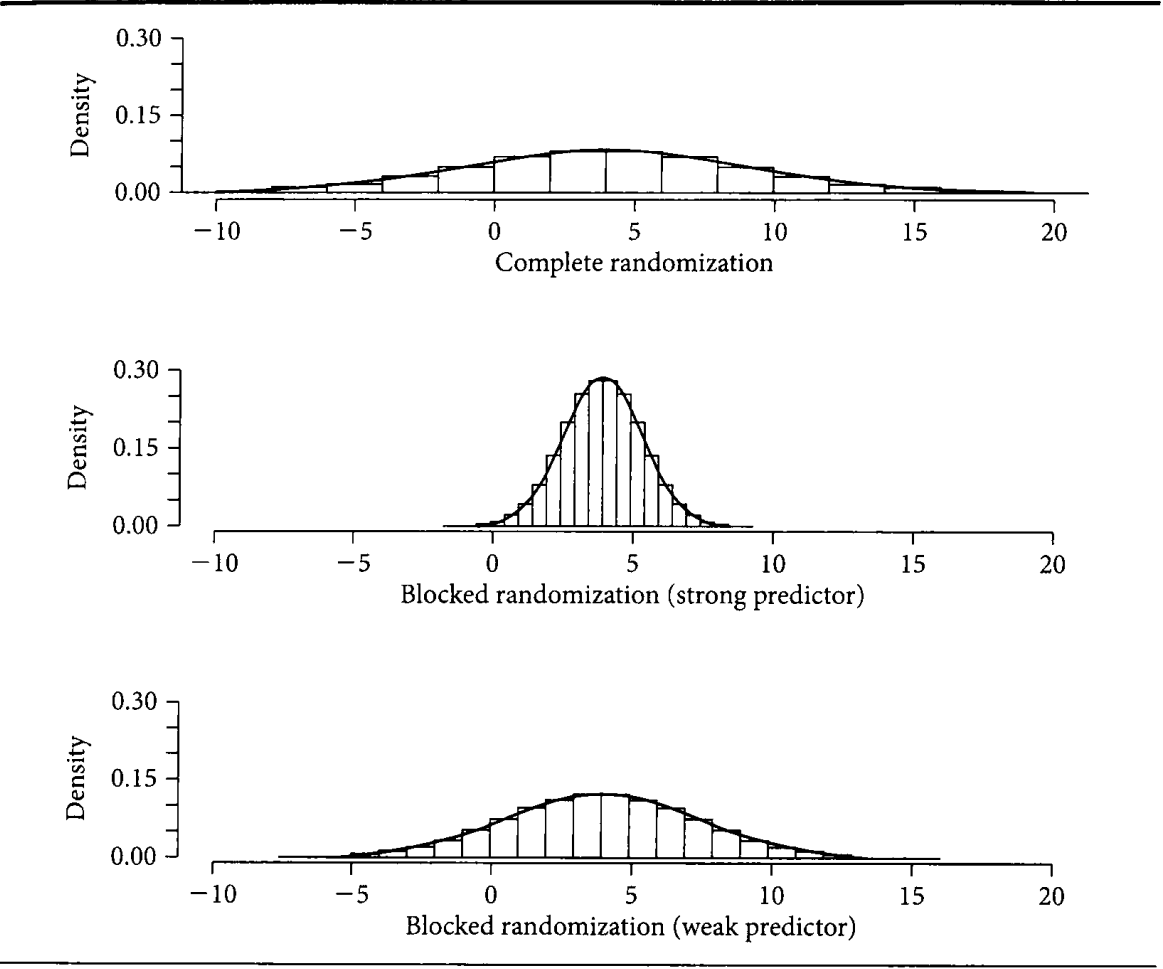
The statistical advantages of blocking depend on the prognostic quality of the covariates used to generate blocks. In this example, blocking results in a substantial improvement in precision because the pre-test strongly predicts outcomes. When the difference-in-means estimator is applied to block randomized data, the estimated ATE is 4.0 with a standard error of 1.35. If the covariate is a weak predictor of outcomes, blocking will do less to improve precision. For example, if we were to repeat the blocking exercise using the noisy pre-test as the basis for forming blocks, the improvement in precision is much less dramatic: the standard error is 3.18.

Figure 4.2 compares the sampling distributions of the difference-in-means estimator across three experimental designs: complete random assignment, blocked

16 Bruhn and McKenzie 2009. Another potential downside of blocking is that it makes it easier for those implementing an experiment to anticipate and subvert the treatment assignment; see Hewitt and Torgerson 2006.

FIGURE 4.2

Comparison of sampling distributions based on completely randomized and block randomized designs



random assignment using the strongly prognostic pre-test, and blocked random assignment using the noisy pre-test. The graph illustrates the principal advantage of blocking. By excluding many of the randomizations that produce outlandish results, blocking narrows the sampling distribution. The advantages of blocking diminish when less prognostic covariates are used to form blocks. Although the noisy pre-test measure predicts 50% of the variance in $Y_i(0)$, many observations that have the same noisy pre-test score turn out to have very different potential outcomes. If an even weaker pre-test were administered such that the scores were nothing but random noise, blocked randomization would generate as much sampling variability as complete randomization.

In terms of precision, the advantages of blocking are less dramatic when the alternative is covariate adjustment using regression. Like blocking, covariate adjustment controls for differences in pre-test scores. Blocking's advantage over controlling for covariates using regression derives from the fact that blocking eliminates randomly

occurring correlation between the treatment and the covariate used to form blocks. The smaller the sample, the bigger blocking's advantage will be. For experiments with more than 100 observations in both the treatment and control groups, there are seldom appreciable differences in terms of precision between block randomized designs and complete randomized designs that are analyzed using regression with controls for covariates.¹⁷

In our running example, we have just 20 observations in each experimental group, and blocking's advantages are noticeable. If, instead of blocking, we had conducted a complete randomization but controlled for each pair of sorted pre-test scores using regression, the estimated treatment effect would average 4.00 with a standard error of 2.05. Due to the collinearity penalty described above, covariate adjustment is not as precise as blocking, which generates estimates whose standard error is 1.35. Nevertheless, covariate adjustment is a dramatic improvement over the difference-in-means estimator, which has a standard error of 4.77.

The practical implications of this discussion may be summarized as follows:

1. Blocking is a useful design tool for reducing sampling variability when the researcher has access to covariates that are thought to predict outcomes. When difference-in-means estimation is used, blocked randomization tends to lead to substantially more precise estimates than complete randomization. Blocking's advantage over regression-based covariate adjustment tends to be modest in samples with more than 100 observations. In small samples, especially when $N < 20$, blocking has two advantages over covariate adjustment: blocking improves precision, and the difference-in-means estimator remains unbiased, whereas regression may be biased.
2. Blocking has the advantage of making transparent the researcher's prior beliefs about which covariates are most likely to predict outcomes. The manner in which blocks are formed provides a clear template for analyzing the experiment's results.
3. By eliminating the correlation between the blocked covariates and treatment assignment, blocking sidesteps the problems of interpretation that arise when adjusted and unadjusted estimates differ.
4. When using randomization inference to test the sharp null hypothesis on block-randomized data, the simulated randomizations must occur within blocks. When simulating randomizations, always follow the procedure that was originally implemented.

17 Rosenberger and Lachin 2002. Some calculations using the binomial distribution illustrate how covariate imbalance and the resulting collinearity penalty tend to diminish with sample size. Recalling our previous example of 10 male and 10 female subjects, half of whom are randomly assigned to treatment, there is a 0.503 probability that random assignment creates a treatment group with 8 or fewer men or 12 or more men. If our sample size increases to 100 men and 100 women, there is just a 0.006 probability that the treatment group will contain 80 or fewer men or 120 or more men.

5. Precision gains from blocking are limited when one's blocks do not differ appreciably in terms of their average potential outcomes. When the covariates used to form blocks have no predictive value whatsoever, the sampling distribution of the estimated ATE under blocked random assignment is approximately the same as under complete random assignment.

BOX 4.3**Randomization-based Inference versus Large-sample Approximations**

Chapters 3 and 4 present a randomization-based framework for testing hypotheses and constructing confidence intervals. The three attractive features of this approach are that (1) it grows directly out of the random allocation procedure and is easily tailored to designs that are blocked or clustered, (2) it may be adapted to a wide array of estimation problems, and (3) when calculating critical values, it does not rely on assumptions about the shape of the sampling distribution.

Notwithstanding the advantages of randomization inference, relatively few social scientists currently use randomization-based methods, preferring instead to test hypotheses and construct confidence intervals using a normal approximation based on the standard errors reported by regression. Under simple or complete randomization, the numbers reported by regression software are frequently quite close to what one would obtain using randomization-based methods. When samples are large or when the disturbances in a regression equation are drawn from a normal distribution, regression tends to generate hypothesis tests and confidence intervals that match what one obtains using randomization inference. When neither condition holds, regression's assumptions about the shape of the sampling distribution may produce misleading results. Under these same conditions, one also finds a close correspondence between randomization inference and regression-based confidence intervals when weighted least squares is applied to a block randomized design. Under cluster assignment, regression generates similar hypothesis tests and confidence intervals to randomization inference so long as the number of clusters is large and one instructs regression software to calculate "robust cluster" standard errors, which assume that disturbances are independent across clusters but possibly correlated within clusters. In sum, regression offers a convenient way to test hypotheses and construct confidence intervals under the assumption of constant treatment effects. The assumptions that regression makes about the shape of the sampling distribution may break down, however, especially when the number of observations (or clusters) is small.

4.5 Analysis of Block Randomized Experiments with Treatment Probabilities That Vary by Block

When blocking the teacher incentives experiment, we implemented a blocking scheme in which one member of each pair of schools was assigned to the treatment group. This design implies that the probability of being assigned to the treatment group was identical from pair to pair. The same would be true if we had created four blocks of five schools each, randomly assigning three members of each block to the treatment group. When treatment probability is identical across blocks, estimation is straightforward: compare the average outcome in the treatment group to the average outcome in the control group.

As stressed in Chapter 3, estimation becomes more complicated when the probability of being assigned to the treatment varies from block to block. For example, suppose we were to divide the 40 schools in the teacher incentives experiment into two blocks based on their pre-test scores. The 24 schools with the lowest pre-test scores form one block, and the remaining 16 schools form the other block. Suppose we were to randomly assign 16 of the low-scoring schools to the treatment and the remaining 8 to control. Among high-scoring schools, we randomly allocate 4 schools to the treatment and the remaining 12 to control. In the first block, the probability of treatment is $2/3$, whereas in the second block this probability is $1/4$. If we simply compare the treatment and control groups without controlling for block, our estimator will be biased. The 20 schools in the treatment group have an expected outcome of 29.2. The 20 schools in the control group have an expected outcome of 35.3. Despite the fact that the true ATE is 4.0, the estimated ATE produced by a difference-in-means comparison is, on average, -6.1 ! Pooling observations across blocks leads to severe bias in this example because treatment assignment is correlated with test scores. By design, schools with higher scores are less likely to receive the treatment.

Random assignment of subjects occurred within blocks, and the appropriate way to estimate the ATE is to compare subjects within each block, not across blocks. The ATE among the 24 schools in the low-achieving block is 3.625. The ATE among the 16 schools in the high-achieving block is 4.5625. In order to obtain the overall ATE, we calculate a weighted average of the two ATEs, with weights equal to each block's share of the overall sample. When estimating the sample average treatment effect using actual data, one uses the estimator presented in equation (3.10). Suppose the estimated ATEs in each block were equal to 3.625 and 4.5625:

$$\left(\frac{N_{Low}}{N}\right)\widehat{ATE}_{Low} + \left(\frac{N_{High}}{N}\right)\widehat{ATE}_{High} = \left(\frac{24}{40}\right)3.625 + \left(\frac{16}{40}\right)4.5625 = 4. \quad (4.11)$$

The estimator in equation (3.10) can be implemented using regression, applying weights to each observation. Each treated observation is weighted by the inverse of

the proportion of subjects in its block who were assigned to the treatment condition, and each control subject is weighted by the inverse of the proportion of subjects in its block who were assigned to the control condition. In order to express these weights compactly, define $p_{ij} \equiv m_j/N_j$ as the probability that subject i in block j is assigned to treatment, and weight each observation by:

$$w_{ij} \equiv \left(\frac{1}{p_{ij}}\right)d_i + \left(\frac{1}{1-p_{ij}}\right)(1 - d_i) = \frac{d_i}{p_{ij}} + \frac{1-d_i}{1-p_{ij}}. \quad (4.12)$$

Most regression software makes it easy for users to impose weights such as w_{ij} . See exercise 9 for a detailed example.

One reason to use regression in order to implement the estimator in equation (3.10) is that one can easily augment the model to control for additional covariates. For example, a researcher may block schools based on information that is readily available at the time of random assignment and then control for covariates that become available afterward. Returning to the stylized example presented in Table 4.1, suppose a researcher had blocked the schools according to whether the student body is predominantly Hindu, Muslim, or Sikh. Random assignment occurs with different probabilities within each block. Later, the researcher obtains pre-test scores for each school. Table 4.2 reports each school's random assignment. Within the predominantly Hindu block, the probability of treatment assignment is 0.6, and so the 10 control group schools are weighted by $1/(1 - 0.6) = 2.5$, and the 15 treatment group schools are weighted by $1/0.6 = 1.67$. Within the predominantly Muslim block, the probability of treatment assignment is 0.5, and so the 5 control group schools are weighted by $1/(1 - 0.5) = 2$, and the 5 treatment group schools are also weighted by $1/0.5 = 2$. Finally, within the predominantly Sikh block, the probability of treatment assignment is 0.4, and so the 3 control group schools are weighted by $1/(1 - 0.4) = 1.67$, and the 2 treatment group schools are weighted by $1/0.4 = 2.5$. Notice that (1) within each block, the weighted N in the treatment group equals the weighted N in the control group (so that the weighted probability of treatment is 0.5), and (2) the sum of the weights in the treatment group is N , and the sum of the weights in the control group is N .

Our regression model for these weighted observations is:

$$Y_i = a + bd_i + cX_i + u_i, \quad (4.13)$$

where d_i is the treatment indicator and X_i is the noisy pre-test (this variable is labeled X_weak_i in Table 4.2). Using the observed outcomes in Table 4.2, weighted least squares regression produces an estimated ATE of 3.925. This relationship can be visualized by creating a scatterplot, shown in Figure 4.3: the X-axis represents the residuals from a weighted regression of d_i on X_i , and the Y-axis represents the residuals from a weighted regression of Y_i on X_i . This plot depicts the weights by varying the size of the hollow circles used to represent each subject. The slope of the regression line is 3.925. Assuming a constant effect of 3.925 in order to form a full schedule

TABLE 4.2

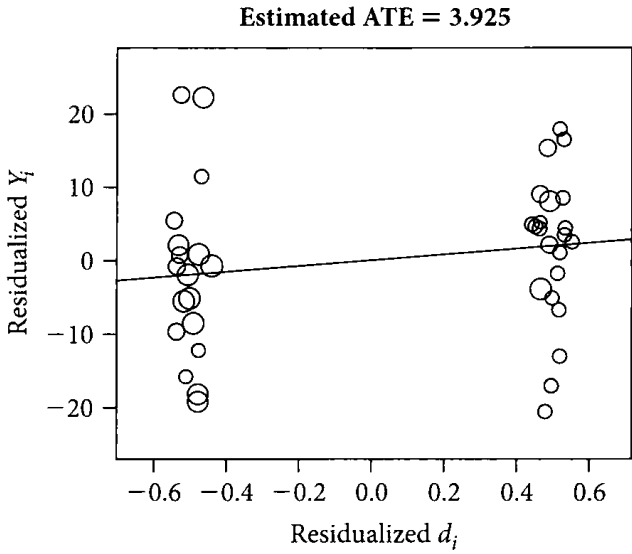
Block random assignment of teacher incentives experiment

Observation	Y_i	d_i	X_weak_i	Block	Probability	Weight*
1	5	0	25	1: Predominantly Hindu	0.6	2.5
2	15	1	12	1: Predominantly Hindu	0.6	1.667
3	6	0	25	1: Predominantly Hindu	0.6	2.5
4	19	1	27	1: Predominantly Hindu	0.6	1.667
5	10	0	10	1: Predominantly Hindu	0.6	2.5
6	11	0	24	3: Predominantly Sikh	0.4	1.667
7	24	1	21	1: Predominantly Hindu	0.6	1.667
8	11	1	25	1: Predominantly Hindu	0.6	1.667
9	16	1	35	1: Predominantly Hindu	0.6	1.667
10	25	1	28	1: Predominantly Hindu	0.6	1.667
11	18	1	41	1: Predominantly Hindu	0.6	1.667
12	20	0	38	3: Predominantly Sikh	0.4	1.667
13	20	0	30	1: Predominantly Hindu	0.6	2.5
14	24	1	20	1: Predominantly Hindu	0.6	1.667
15	24	0	24	1: Predominantly Hindu	0.6	2.5
16	26	1	26	1: Predominantly Hindu	0.6	1.667
17	30	1	22	1: Predominantly Hindu	0.6	1.667
18	27	0	34	1: Predominantly Hindu	0.6	2.5
19	43	1	37	3: Predominantly Sikh	0.4	2.5
20	32	0	21	3: Predominantly Sikh	0.4	1.667
21	32	0	40	1: Predominantly Hindu	0.6	2.5
22	27	1	34	1: Predominantly Hindu	0.6	1.667
23	32	0	36	1: Predominantly Hindu	0.6	2.5
24	37	1	37	2: Predominantly Muslim	0.5	2
25	35	0	48	2: Predominantly Muslim	0.5	2
26	39	1	46	3: Predominantly Sikh	0.4	2.5
27	42	1	25	1: Predominantly Hindu	0.6	1.667
28	37	1	21	1: Predominantly Hindu	0.6	1.667
29	41	0	19	1: Predominantly Hindu	0.6	2.5
30	42	0	44	2: Predominantly Muslim	0.5	2
31	51	1	50	2: Predominantly Muslim	0.5	2
32	44	0	48	2: Predominantly Muslim	0.5	2
33	45	0	46	1: Predominantly Hindu	0.6	2.5
34	49	1	47	1: Predominantly Hindu	0.6	1.667
35	48	1	47	1: Predominantly Hindu	0.6	1.667
36	52	1	39	2: Predominantly Muslim	0.5	2
37	52	0	50	2: Predominantly Muslim	0.5	2
38	52	1	46	2: Predominantly Muslim	0.5	2
39	55	1	54	2: Predominantly Muslim	0.5	2
40	62	0	42	2: Predominantly Muslim	0.5	2

* Weight will vary with the randomization. $Pr(D = 1 | Block = 1) = 15/25 = 0.6$; $Pr(D = 1 | Block = 2) = 5/10 = 0.5$; $Pr(D = 1 | Block = 3) = 2/5 = 0.4$.

FIGURE 4.3

Scatterplot of residuals from weighted regressions of d_i on X_weak_i and Y_i on X_weak_i . Estimated ATE = 3.925; plotted circles are sized to reflect each observation's weight.



of potential outcomes and simulating 100,000 random assignments, we obtain a 95% confidence interval for the ATE that ranges from -2.75 to 10.67 . When we generate the full schedule of potential outcomes assuming the sharp null hypothesis of no treatment effect, we find that 13.3% of the simulated random assignments are larger than 3.925, implying a p -value of 0.13.

This weighting method raises a subtle point about the use of regression to estimate the ATE when experiments employ blocking. A common estimation approach is to use unweighted regression: researchers often regress outcomes on the treatment, controlling for a series of dummy variables that indicate each block. This unweighted regression can be shown to be equivalent to a weighted regression in which each block's estimated ATE is weighted by $(N_j/N)P_j(1 - P_j)$, where N_j/N refers to the share of the total sample that is in block j , and where P_j refers to the proportion of subjects in block j that are assigned to the treatment group.¹⁸ When the probability of assignment to the treatment group is such that $P_j(1 - P_j)$ is constant across blocks, regression and the weighted average estimator in equation (3.10) give identical results. When this condition does not hold, the two estimators may give different results. In our example, the $P_j(1 - P_j)$ for the three blocks are similar ($6/25$ for blocks 1 and 3, and $1/4$ for block 2), so, on average, regression produces an estimate of 3.96, which is close to the ATE. Regression with dummy variables makes sense

18 Each of the weights is normalized by dividing by the sum of all weights; the sum of the normalized weights is therefore 1. See Humphreys 2009 for derivation and illustration of this point.

BOX 4.4**Displaying Covariate-Adjusted Estimates of the ATE under Simple or Complete Random Assignment**

In order to depict regression estimates of the ATE when the regression model includes covariates, perform the following steps:

1. Regress d_i on *all* covariates and compute the residual for each observation. Call these residuals e_{di} .
2. Regress Y_i on the covariates (do not include d_i) and compute the residual for each observation. Call these residuals e_{yi} .
3. Plot the values of e_{di} and e_{yi} , placing e_{di} on the X-axis and e_{yi} on the Y-axis.

Verify that you have followed the steps correctly by regressing e_{yi} on e_{di} ; this regression should generate the same estimated ATE as a regression of Y_i on d_i and the covariates.

Displaying Covariate-Adjusted Estimates of the ATE under Block Random Assignment

Follow the same three steps, but in steps (1) and (2), perform a weighted regression using the weights in equation (4.12). In step (3), plot each observation using a hollow circle that is sized to reflect each observation's weight. See Figure 4.3 and the online appendix at <http://isps.research.yale.edu/FEDAI> for annotated examples.

when you are prepared to assume that treatment effects are the same for all subjects; in that case, you want to weight each block by the precision with which the treatment effect is estimated.

Regardless of whether one controls for blocks using weighted regression or regression with indicators for blocks, the key principle is to compare treatment and control subjects within, not between, blocks. The same principle applies to the presentation of statistics describing covariate balance when the probability of treatment assignment varies by block; presenting treatment and control averages only makes sense *within* each block. In other words, when presenting statistical descriptions of the treatment and control groups, subdivide the table so that each block's results are presented separately.¹⁹ This principle also applies to figures that illustrate the distribution of

19 If the number of blocks is so numerous that block-by-block tables become cumbersome, weight each observation by w_{ij} as defined in equation (4.12) and, for the treatment and control groups, present the weighted means for each covariate.

BOX 4.5**Restricted Random Assignment**

Sometimes the time constraints under which field experiments are conducted make blocking impossible. When using complete random assignment, a researcher may quickly approximate blocking by conducting a test of balance and redoing the randomization if the balance criterion is not satisfied (e.g., one obtains a p -value below 0.05 when using the F-statistic from a regression to test the null hypothesis that none of the covariates predict treatment assignment). This “catch-and-release” rule results in randomly assigned groups that have similar covariate profiles.

When using restricted randomization, remember to make two adjustments when analyzing the results of your experiment. First, if your treatment and control groups are different sizes, your randomization procedure may cause the probability of being assigned to the treatment group to vary from one subject to the next. For example, subjects with very unusual covariate profiles may be less likely to be assigned to the smaller of the two experimental groups, because their presence in that assigned group would cause a potential random assignment to fail the balance test. In order to obtain unbiased estimates of the ATE, you must weight the observations by w_i , the inverse of the probability of assignment to the observed experimental group. These weights are estimated by simulating a large number of random assignments and tabulating the fraction of random assignments that assign each subject to treatment or control. Second, when conducting randomization inference, you must throw out any random allocations that would have failed your balance test. It is important to document this rejection rule in detail so that randomization tests can exactly reproduce the set of admissible randomizations. See exercise 4.5 for an example of how to estimate the ATE and conduct hypothesis tests under restricted randomization.

outcomes in the treatment and control groups. If the number of blocks is small, these figures should depict relationships for each block separately. If you have a large number of blocks, weight the observations using equation (4.12), and depict more heavily weighted observations with larger symbols.

SUMMARY

By making judicious use of covariates, experimental researchers can improve the precision with which average treatment effects are estimated. Sometimes the improvement is dramatic, equivalent to a marked increase in the number of observations.

The challenge is to use covariates in a principled way, because any gain in precision can be offset by procedures that potentially introduce bias. Covariates should be chosen from the set of variables that exist prior to the random intervention. Ideally, the particular covariates that the researcher wishes to control should be specified in advance of seeing the experimental outcomes; otherwise, the selective use of covariates raises concerns about whether the researcher is trying to guide the estimates toward a particular result. Block randomized designs are particularly appealing in this regard. Although blocking often produces a relatively small gain in precision over after-the-fact regression adjustment of completely randomized experiments, blocking has the virtue of embodying *ex ante* expectations about prognostic variables and a specific model for estimating treatment effects. If the probability of treatment assignment varies by block, be sure to take this aspect of the design into account when analyzing the data; comparing unweighted outcomes in treatment and control groups may produce severely biased estimates if treatment probabilities are correlated with potential outcomes.

Covariates are of special value to researchers who must conduct experiments under severe budget constraints. Squeezing every drop of precision out of an experiment requires a bit of extra effort and ingenuity. Researchers should be on the lookout for target populations for which extensive background information is readily available. The Swedish government, for example, makes available to researchers extraordinarily detailed information about its citizens' school records, employment history, and health records. Researchers conducting surveys or social network mapping may have rich data on individuals who could later be the subjects of an experimental intervention. Firms that distribute credit or discount cards sometimes have extensive information about the purchasing behavior of their customers. Experiments that are built on these databases start with a wealth of prognostic covariates.²⁰

Covariates also assist in the detection of administrative errors. Mistakes are inevitable in the collection and processing of primary data, and one virtue of random assignment is that it creates well-defined statistical patterns that can be detected with the aid of covariates. No method of error detection is foolproof, but when covariates are strongly related to treatment assignment, researchers should review the procedures used to allocate observations to experimental groups.

One danger in mixing covariates into the analysis of experimental data is the temptation to interpret covariates in causal terms. It is not uncommon to encounter research articles in which authors offer a causal interpretation for all of the regression coefficients they present, including the coefficients associated with pre-test scores, blocking indicators, and other factors that are not subject to random assignment.

20 These covariates may also prove valuable in the detection of heterogeneous treatment effects, a topic to which we return in Chapter 9.

Attaching a causal story to these coefficients is at odds with the experimental approach, which uses random assignment to build up a stock of defensible causal claims. The coefficients associated with covariates are uninterpretable without strong substantive assumptions. Predominantly Hindu schools might perform better on average than Muslim schools due to Hindu childrearing practices, or it could just be that religion is a marker for a host of unmeasured attributes that are correlated with educational outcomes. The set of covariates included in an experimental analysis need not be a complete list of factors that affect outcomes; the fact that some factors are left out or are poorly measured is not a source of bias when the aim is to measure the average treatment effect of the random intervention. Omitted variables and mismeasurement, however, can lead to severe bias if the aim is to draw causal inferences about the effects of covariates. Causal interpretation of the covariates encounters all of the threats to inference associated with analysis of observational data.

SUGGESTED READINGS

For an accessible introduction to the mechanics of regression, see Freedman, Pisani, and Purves (2007) and, for more technical detail, Angrist and Pischke (2009). The use of regression to adjust for covariates is critiqued in Freedman (2008), but Schochet (2010) and Green and Aronow (2011) suggest that regression tends to produce approximately unbiased estimates unless samples are very small. Randomization inference for checking balance is discussed in Hansen and Bowers (2008). Lock (2011) discusses the properties and drawbacks of restricted randomization. Humphreys (2009) discusses the use of unweighted regression to analyze block randomized experiments with assignment probabilities that vary by block. Wooldridge (2002) discusses methods for correcting regression standard errors to account for clustering.

EXERCISES: CHAPTER 4

See <http://isps.research.yale.edu/FEDAI> for datasets and programs for conducting randomization inference.

1. Important concepts:
 - (a) Define “covariate.” Explain why covariates are (at least in principle) measured prior to the random allocation of subjects to treatment and control.
 - (b) Define “disturbance term.”
 - (c) In equation (4.2), we demonstrated that rescaling the outcome by subtracting a pre-test leads to unbiased estimates of the ATE. Suppose that instead of subtracting the pre-test X_i , we subtracted a rescaled pretest cX_i , where c is some positive constant. Show that this procedure produces unbiased estimates of the ATE.
 - (d) Show that the parameter b in equation (4.7) is identical to the ATE.
2. A researcher working with Israeli elementary school students sought to improve students’ ability to solve logic puzzles.²¹ Students in the treatment and control group initially took

21 Dan Gendelman conducted this study in 2004 and shared it with us via personal communication.

a computer-administered test, and the number of correctly solved puzzles was recorded. A few days later, students assigned to the control group were then given 30 minutes to improve their puzzle-solving skills by playing on a computer. During the same allotment of time, students in the treatment group listened to an instructor describe some rules of thumb to keep in mind when solving logic puzzles. All subjects then took a computer-administered post-test, and the number of correctly solved puzzles was recorded. The table below shows the results for each subject.

Subject	d_i	Pre-test	Post-test	Improvement
1	1	10	10	0
2	1	9	11	2
3	1	5	6	1
4	1	3	6	3
5	1	3	6	3
6	1	6	7	1
7	1	6	7	1
8	1	5	6	1
9	1	6	7	1
10	0	9	9	0
11	0	6	7	1
12	0	11	10	-1
13	0	4	5	1
14	0	3	3	0
15	0	10	10	0
16	0	7	8	1
17	0	7	7	0
18	0	8	10	2

- (a) As a randomization check, use randomization inference to test the null hypothesis that the pre-test scores are unaffected by treatment assignment.
 - (b) Use difference-in-means estimation to estimate the effect of the treatment on the post-test score. Form a 95% confidence interval.
 - (c) Use difference-in-differences estimation to estimate the effect of the treatment on the post-test score. Form a 95% confidence interval, and compare it to the interval in part (b).
3. The table below illustrates the problems that may arise when researchers exercise discretion over what results to report to readers. Suppose the true ATE associated with a given treatment were 1.0. The table reports the estimated ATE from nine experiments, each of which involves approximately 200 subjects. Each study produces two estimates, one based on a difference-in-means and another using regression to control for covariates. In principle, both estimators generate unbiased estimates, and covariate adjustment has a slight edge in terms of precision. Suppose the researchers conducting each study use the following decision rule: “Estimate the ATE using both estimators and report whichever estimate is larger.” Under this reporting policy, are the reported estimates unbiased? Why or why not?

Study	No covariates	With covariates
1	5	4
2	3	3
3	2	2
4	6	5
5	1	1
6	0	0
7	−3	−1
8	−5	−4
9	0	−1
Average	1.00	1.00
Standard Deviation	3.54	2.83

4. Table 4.1 contains a column of treatment assignments, d_i , that reflects a complete random assignment of 20 schools to treatment and 20 schools to control.
- (a) Use equation (2.2) to generate observed outcomes based on these assigned treatments. Regress Y_i on d_i and interpret the slope and intercept. Is the estimated slope the same as the estimated ATE based on a difference-in-means?
 - (b) Regress treated and untreated outcomes on X_i to see whether the condition in equation (4.6) appears to hold. What do you infer about the advisability of rescaling the dependent variable so that the outcome is a change score (i.e., $Y_i - X_i$)?
 - (c) Regress Y_i on d_i and X_i . Interpret the regression coefficients, contrasting these results with those obtained from a regression of Y_i on d_i alone.
 - (d) With the estimates obtained in part (a), use randomization inference (as described in Chapter 3) to evaluate the sharp null hypothesis of no effect for any school. To obtain the sampling distribution under the sharp null hypothesis, simulate 100,000 random assignments, and for each simulated sample, estimate the ATE using a regression of Y_i on d_i . Interpret the results.
 - (e) With the estimates obtained in part (c), use randomization inference to evaluate the sharp null hypothesis of no effect for any school. To obtain the sampling distribution under the sharp null hypothesis, simulate 100,000 random assignments, and for each simulated sample, estimate the ATE using a regression of Y_i on d_i and X_i . Interpret the results.
 - (f) Use the estimated ATE in part (a) to construct a full schedule of potential outcomes for all schools, assuming that every school has the same treatment effect. Using this simulated schedule of potential outcomes, construct a 95% confidence interval for the sample average treatment effect in the following way. First, randomly assign each subject to treatment or control, and estimate the ATE by a regression of Y_i on d_i . Repeat this procedure until you have 100,000 estimates of the ATE. Order the estimates from smallest to largest. The 2,500th estimate marks the 2.5th percentile, and the 97,501st estimate marks the 97.5th percentile. Interpret the results.
 - (g) Use the estimated ATE in part (c) to construct a full schedule of potential outcomes for all schools, assuming that every school has the same treatment effect. Using this

simulated schedule of potential outcomes, simulate the 95% confidence interval for the sample average treatment effect estimated by a regression of Y_i on d_i and X_i . Interpret the results. Is this confidence interval narrower than one you generated in response to question (f)?

5. Randomizations are said to be “restricted” when the set of all possible random allocations is narrowed to exclude allocations that have inadequate covariate balance. Suppose, for example, that the assignment of treatments (d_i) in Table 4.1 was conducted subject to the restriction that a regression of d_i on X_i (the pre-test) generates an F-statistic whose reported p -value is greater than 0.05. In other words, had the researcher found that the assigned d_i were significantly predicted by X_i , the random allocation would have been conducted again, until the d_i met this criterion.
 - (a) Conduct a series of random assignments in order to calculate the weighting variable w_i ; for units in the treatment group, this weight is defined as the inverse of the probability of being assigned to treatment, and for units in the control group, this weight is defined as the inverse of the probability of being assigned to control. See Table 4.2 for an example. Does w_i appear to vary within the treatment group or within the control group?
 - (b) Use randomization inference to test the sharp null hypothesis that d_i has no effect on Y_i by regressing Y_i on d_i and comparing the estimate to the sampling distribution under the null hypothesis. Make sure that your sampling distribution includes only random allocations that satisfy the restriction mentioned above. If the probability of treatment varies from one subject to the next, estimate the ATE by weighting each observation by w_i . Calculate the p -value and interpret the results.
 - (c) Use randomization inference to test the sharp null hypothesis that d_i has no effect on Y_i by regressing Y_i on d_i and X_i (weighting the observations by w_i , if necessary) and comparing the estimate to the sampling distribution under the null hypothesis. Calculate the p -value, and interpret the results.
 - (d) Compare the sampling distributions under the null hypothesis in parts (b) and (c) to the sampling distributions obtained in exercises 4(d) and 4(e), which assumed that the randomization was unrestricted.
6. One way to practice your experimental design skills is to undertake a mock randomization of an existing nonexperimental dataset. In this exercise, the existing dataset is treated as though it were a baseline data collection effort that an experimental researcher gathered in preparation for a random intervention. The actual data in question come from a panel study of Russian villagers.²² Villagers from randomly selected rural areas of Russia were interviewed in 1995 and re-interviewed in 1996 and 1997. Our attention focuses on the 462 respondents who were interviewed in all three waves and provided answers to questions about their income, church membership, and evaluation of national conditions (i.e., how well are things going in Russia?). Imagine that an experimental intervention occurred after the 1996 survey and that national evaluations in the 1997 survey were the experimental outcome of interest. The dataset provided at <http://isps.research.yale.edu/FEDAI> contains the following pre-treatment covariates that may be used for blocking:

22 O'Brien and Patsiorkovski 1999.

sex, church membership, social class, and evaluations of national conditions in 1995 and 1996. As you design your experiment, imagine that “post-intervention” evaluations of national conditions in 1997 were unknown.

- (a) One way to develop a sense of which variables are likely to predict post-intervention evaluations of national conditions in 1997 is to regress evaluations of national conditions in 1996 on sex, church membership, social class, and evaluations in 1995. Which of these variables seem to most strongly predict evaluations of national conditions in 1996? What is the R^2 from this regression?
 - (b) Suppose you were to design a block random assignment in order to predict evaluations in 1997. Use the R package `blockTools` (for example code, see <http://isps.research.yale.edu/FEDAI>) to perform a block random assignment, blocking on sex, church membership, social class, and evaluations in 1996. Decide for yourself how many subjects to include in each block. Compare the treatment and control groups to verify that blocking produced groups that have the same profile of sex, church membership, social class, and evaluations in 1996.
 - (c) Suppose you wanted to assess how well your blocking design performed in terms of increasing the precision with which treatment effects are estimated. Of course, there was no actual treatment in this case, but imagine that shortly after the survey in 1996, a treatment were administered at random to half of the respondents. (Here is an instance in which the sharp null hypothesis of no effect is known to be true!) The outcome from this imaginary experiment is evaluations of national conditions in 1997. Compare the sampling distribution of the estimated treatment effect (which should be centered on zero) under complete random assignment to the sampling distribution of the estimated treatment effect under block random assignment.
 - (d) Calculate the sampling distribution of the estimated treatment effect under complete random assignment using regression to control for the variables that would have otherwise been used to form blocks. Compare the resulting distribution to the sampling distribution of the estimated treatment effect under block random assignment. Does blocking produce an appreciable gain in precision over what is achieved by covariate adjustment?
7. Researchers may be concerned about using block randomization when they are unsure whether the variable used to form the blocks actually predicts the outcome. Consider the case in which blocks are formed randomly—in other words, the variable used to form the blocks has no prognostic value whatsoever. Below is a schedule of potential outcomes for four observations.
- (a) Suppose you were to use complete random assignment such that $m = 2$ units are assigned to treatment. What is the sampling variance of the difference-in-means estimator across all six possible random assignments?
 - (b) Suppose you were to form blocks by randomly pairing the observations. Within each pair, you randomly allocate one subject to treatment and the other to control so that $m = 2$ units are assigned to treatment. There are three possible blocking schemes; for each blocking scheme, there are four possible random assignments. What is the sampling variance of the difference-in-means estimator across all twelve possible random assignments?

- (c) From this example, what do you infer about the risks of blocking on a non-prognostic covariate?

Subject	Y(0)	Y(1)
A	1	2
B	0	3
C	2	2
D	5	5

8. Sometimes researchers randomly assign subjects from lists that are later discovered to have duplicate entries. Suppose, for example, that a fundraising experiment randomly assigns 500 of 1,000 names to a treatment that consists of an invitation to contribute to a charitable cause. However, it is later discovered that 600 names appear once and 200 names appear twice. Before the invitations are mailed, duplicate invitations are discarded, so that no one receives more than one invitation.
- What is the probability of assignment to the treatment group among those whose names appeared once in the original list? What is the probability of assignment to the treatment group among those whose names appeared twice in the original list?
 - Of the 800 unique names in the original list, how many would you expect to be assigned to treatment and control?
 - What estimation procedure should one use in order to obtain unbiased estimates of the ATE?
9. Gerber and Green conducted a mobilization experiment in which calls from a large commercial phone bank urged voters in Iowa and Michigan to vote in the November 2002 election.²³ The randomization was conducted within four blocks: uncompetitive congressional districts in Iowa, competitive congressional districts in Iowa, uncompetitive congressional districts in Michigan, and competitive congressional districts in Michigan. Table 4.3 presents results only for one-voter households in order to sidestep the complications of cluster assignment.
- Within each of the four blocks, what was the apparent effect of being called by a phone bank on voter turnout?
 - When all of the subjects in this experiment are combined (see the rightmost column of the table), turnout seems substantially higher in the treatment group than the control group. Explain why this comparison gives a biased estimate of the ATE.
 - Using the weighted estimator described in Chapter 3, show the calculations used to provide an unbiased estimate of the overall ATE.
 - When analyzing block randomized experiments, researchers frequently use regression to estimate the ATE by regressing the outcome on the treatment and indicator variables for each of the blocks (omitting one indicator variable if the regression includes an intercept.) This regression estimator places extra weight on blocks that allocate approximately half of the subjects to the treatment condition (i.e., $P_j = 0.5$)

23 Gerber and Green 2005.

Data from the Iowa and Michigan voter mobilization experiment

	BLOCK #1: Uncompetitive Iowa congressional districts		BLOCK #2: Competitive Iowa congressional districts		BLOCK #3: Uncompetitive Michigan congressional districts		BLOCK #4: Competitive Michigan congressional districts		Total	
	Control	Treatment	Control	Treatment	Control	Treatment	Control	Treatment	Control	Treatment
% Voting	48.34	49.30	56.57	55.79	43.62	42.26	48.24	49.07	46.54	48.97
N	39,618	6,935	136,074	7,007	581,919	7,548	154,385	7,229	911,996	28,719
Total N	46,553		143,081		589,467		161,614		940,715	
Share of total N in each block (W_j)	0.049487		0.1520981		0.626616		0.171799			
Estimated ATE	0.00964		-0.007829		-0.01362		0.008271			
Estimated SE	0.006505		0.0060725		0.005745		0.006013			
Weighted average of estimated ATEs with weights equal to W_j									-0.007828	
Standard error (square root of weighted average of estimated variances with weights equal to $W_j^2 / \sum_j W_j^2$)									0.00387071	
Proportion of subjects assigned to treatment (P_j)	0.14897		0.0489723		0.012805		0.04473			
Weights used by OLS: $W_j P_j (1 - P_j) / \sum (W_j P_j (1 - P_j))$	0.219216		0.2475177		0.276768		0.256498			
Weighted average of estimated ATEs with weights equal to $W_j P_j (1 - P_j) / \sum (W_j P_j (1 - P_j))$									-0.001472823	
OLS estimate, controlling for blocks (standard error in parentheses)									-0.0014728 (.0030273)	

because these blocks tend to estimate the within-block ATE with less sampling variability. Compare the four OLS weights to the weights w_i used in part (c).

- (e) Regression provides an easy way to calculate the weighted estimate of the ATE in part (c) above. For each treatment subject i , compute the proportion of subjects in the same block who were assigned to the treatment group. For control subjects, compute the proportion of subjects in the same block who were assigned to the control group. Call this variable q_i . Regress outcomes on treatment, weighting each observation by $1/q_i$, and show that this type of weighted regression produces the same estimate as weighting the estimated ATEs for each block.
10. The 2003 Kansas City voter mobilization experiment described in Chapter 3 is a cluster randomized design in which 28 precincts comprising 9,712 voters were randomly assigned to treatment and control.²⁴ The study contains a wealth of covariates: the registrar recorded whether each voter participated in elections dating back to 1996. The data-set may be obtained at <http://isps.research.yale.edu/FEDAI>.
- (a) Assess the balance of the treatment and control groups by looking at whether past turnout predicts treatment assignment. Regress treatment assignment on the entire set of past votes, and record the associated F-statistic. Use randomization inference to test the null hypothesis that none of the past turnout variables predict treatment assignment. Remember that to simulate the distribution of the F-statistic, you must generate a large number of random *cluster assignments* and calculate the F-statistic for each simulated assignment. Judging from the p -value of this test, what does the F-statistic seem to suggest about whether subjects in the treatment and control groups have comparable background characteristics?
 - (b) Regress turnout in 2003 (after the treatment was administered) on the experimental assignment and the full set of covariates. Interpret the estimated ATE. Use randomization inference to test the sharp null hypothesis that experimental assignment had no effect on any subject's decision to vote.
 - (c) When analyzing cluster randomized experiments with clusters of varying size, one concern is that difference-in-means estimation is prone to bias. This concern also applies to regression. In order to sidestep this problem, researchers may choose to use the difference-in-totals estimator in equation (3.24) to estimate the ATE. Estimate the ATE using this estimator.
 - (d) Use randomization inference to test the sharp null hypothesis that treatment assignment had no effect, using the difference-in-totals estimator.
 - (e) The difference-in-totals estimator can generate imprecise estimates, but its precision can be improved by incorporating information about covariates. Create a new outcome variable that is the difference between a subject's turnout ($1 = \text{vote}$, $0 = \text{abstain}$) and the average rate of turnout in all past elections. Using this "differenced" outcome variable, estimate the ATE using the difference-in-totals estimator, and test the sharp null hypothesis of no effect.

²⁴ Arceneaux 2005.